

Characterizing the Magnetization of Magnetite (Fe_3O_4) via a Vibrating Sample Magnetometer

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Abstract

The magnetic properties of magnetite (Fe_3O_4) were investigated using a vibrating sample magnetometer (VSM) in a Quantum Design VersaLab system. Measurements of magnetic moment as a function of applied magnetic field, $M(H)$, and temperature, $M(T)$, were performed in order to characterize the ferrimagnetic behavior of the material and identify signatures of the Verwey transition. Hysteresis loops measured at 300 K and 50 K exhibited clear coercivity and remanent magnetization, confirming the ferrimagnetic nature of the magnetite sample. From the $M(H)$ data, the coercive field was determined to be approximately $H_c \approx 540$ Oe and the saturation moment was approximately 0.089 emu. Normalizing by the sample mass yielded an estimated saturation magnetization of about 0.5 emu/g. The $M(T)$ measurements revealed an anomaly near $T_V \approx 119$ K, consistent with the Verwey transition associated with charge ordering and a structural transformation in magnetite.

1 Introduction

Magnetite (Fe_3O_4), historically known as lodestone, is one of the earliest naturally recognized magnetic minerals and was widely sought in the Middle Ages due to its strong ferromagnetic behavior. Today, magnetite is used in a variety of industrial applications including inks and cosmetics. Despite these practical uses, the material remains of significant interest in condensed matter physics because of the Verwey transition.

In 1939, E. J. W. Verwey reported that the electrical resistivity of magnetite increases by roughly two orders of magnitude as the temperature is lowered [1]. This behavior is identified as a charge-ordering phase transition in which the ordering of charge removed the needed fluctuations for conduction [2]. Previous experiments have shown that this transition is coupled by a structural change. At high temperatures, magnetite has a cubic inverse spinel structure, while at lower temperatures it transforms into a monoclinic phase. Because this transition modifies both the molecular symmetry and charge locations, it directly influences the magnetic and electrical properties of the material.

Magnetism in solids originates from the magnetic moments associated with electron spin and orbital angular momentum. In magnetite, these moments arise primarily from the iron ions in the crystal lattice. Exchange interactions between neighboring ions cause groups of spins to align together, forming magnetic domains. Magnetite is classified as a ferrimagnetic material because magnetic moments on different iron sublattices are aligned in opposite directions but do not fully cancel. In the inverse spinel structure of Fe_3O_4 , the magnetic moments of Fe^{3+} ions on tetrahedral sites oppose those on octahedral sites, while the Fe^{2+} ions contribute an additional uncompensated moment. As a result, the material exhibits a net magnetization.

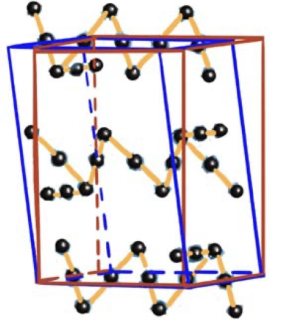


Figure 1: sketch showing the iron sublattice and the structure transition between the higher temperature phase in red and the lower temperature phase spinel in blue

Because the Verwey transition alters both the crystal structure and the electronic ordering of the iron ions, it can influence the magnetic response of the material. Changes in lattice symmetry modify the magnetic structure and domain behavior, which can be detected through measurements of the magnetic moment.

In this experiment, a Quantum Design PPMS (VersaLab) system equipped with a vibrating sample magnetometer (VSM) was used to characterize the magnetic behavior of magnetite. Measurements of magnetic moment as a function of applied magnetic field, $M(H)$, and magnetic moment as a function of temperature, $M(T)$, were performed in order to investigate the magnetization properties of Fe_3O_4 and identify signatures of the Verwey transition.

2 Background

The magnetic moment of the sample was measured using a VSM device, which determines the magnetization by detecting the voltage induced by a changing magnetic flux (Φ). How the VSM does this, is it allows the sample to sit in a uniform magnetic field and let the sample align its own magnetic dipoles with a magnetic moment M . Then the sample is oscillated via a mechanical motor creating a varying moment (μ) and a change in magnetic flux. This magnetic flux (Φ) is then picked up through coils. According to Faraday's law the voltage (V) is proportional to the change of magnetic flux (Φ) [3].

$$V \propto \frac{d\Phi}{dt}$$

Given the magnetic moment the VSM and VersaLab software does a calculation for the magnetization using the equation. Where M is the Magnetism of the sample, μ is the moment determined by the voltage through the coils, and m_{sample} is the mass of the sample.

$$M = \frac{\mu}{m_{\text{sample}}}$$

3 Experimental Setup

Using the Quantum Design PPMS VersaLab with the VSM option [5], we followed the procedures described in the Discovery Teaching Labs Fe_3O_4 Magnetic Characterization experiment [4]. Given that the machine and equipment was brand new and never worked on before in our department by students we followed the instructions closely to the experiment.

VersaLab VSM Measurement Sequence

1. Set temperature to 300 K at 12 K/min.
2. Set magnetic field to 0 Oe and oscillate the magnet.
3. Measure $M(H)$ using a five-quadrant sweep from $-10,000$ to $+10,000$ Oe.
4. Set magnetic field to 1000 Oe.
5. Measure $M(T)$ while cooling from 300 K to 50 K.
6. Set magnetic field to 0 Oe.
7. Set temperature to 50 K.
8. Measure $M(H)$ using a five-quadrant sweep from $-10,000$ to $+10,000$ Oe.
9. Set magnetic field to 1000 Oe.
10. Measure $M(T)$ while warming from 50 K to 300 K.

Figure 2: Measurement sequence used in the VersaLab VSM experiment.

3.1 Sample

The sample of magnetite we chose was one that fit snugly into sample holder, as shown in Fig. 2. Making sure that the sample was 35 mm from the end of the sample holder, a background measurement of the mounting medium was performed to ensure that the holder contribution was small compared to the sample signal. The chamber was purged prior to cooling to reduce contamination from water or oxygen.

3.2 Sequence

Moment-versus-temperature measurements were performed in both zero-field-cooled (ZFC) and field-cooled (FC) modes at an applied field of 1000 Oe. Moment-versus-field measurements were performed using a five-quadrant field sweep to obtain the full hysteresis loop. In our sequence (fig. 2) the software ran both the moment-versus-field, and moment-versus-temperature.

4 Data

The VSM software exports measurement data as text and dat files containing the applied magnetic field, sample temperature, and magnetic moment. The raw files used in this experiment contained columns for temperature (K), magnetic field (Oe), and magnetic moment (emu). These data were used to construct plots of magnetic moment as a function of applied magnetic field $M(H)$ and temperature $M(T)$.

4.1 Postprocessing

The raw data files was processed using a Python script to extract the relevant columns and generate plots of $M(H)$ and $M(T)$. The magnetic moment reported by the VersaLab software is given in units of emu. The magnetization was obtained by normalizing the measured moment by the sample mass of 0.18 g.

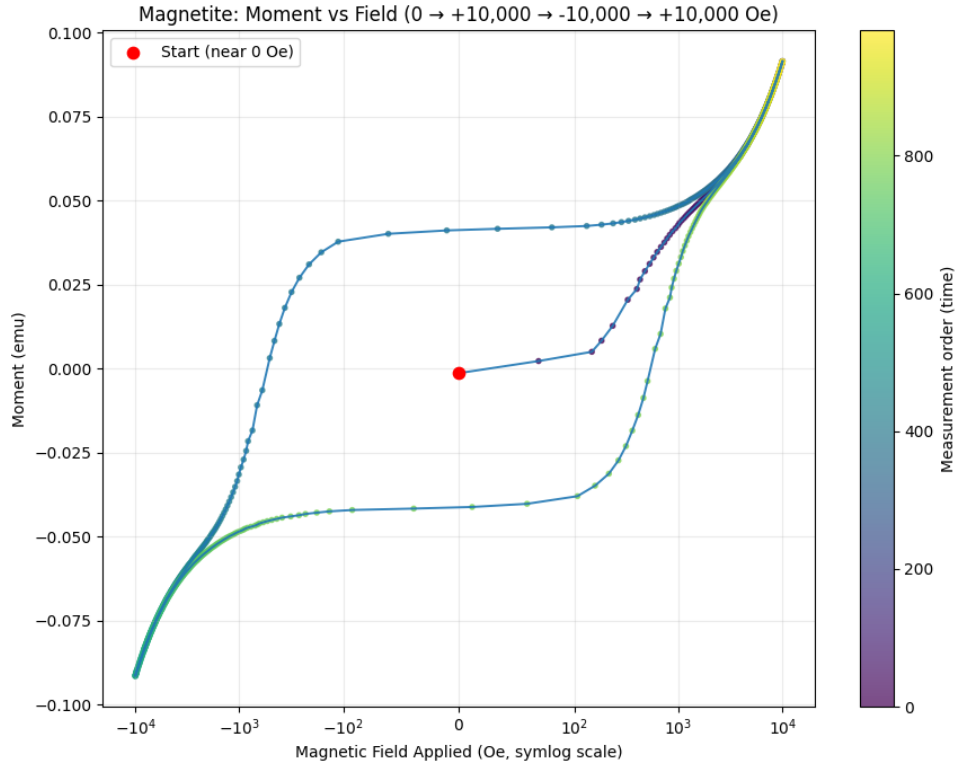


Figure 3: Magnetic moment as a function of applied magnetic field for the magnetite sample at 300 K. The hysteresis loop shows remanent magnetization and coercivity characteristic of ferrimagnetic materials.

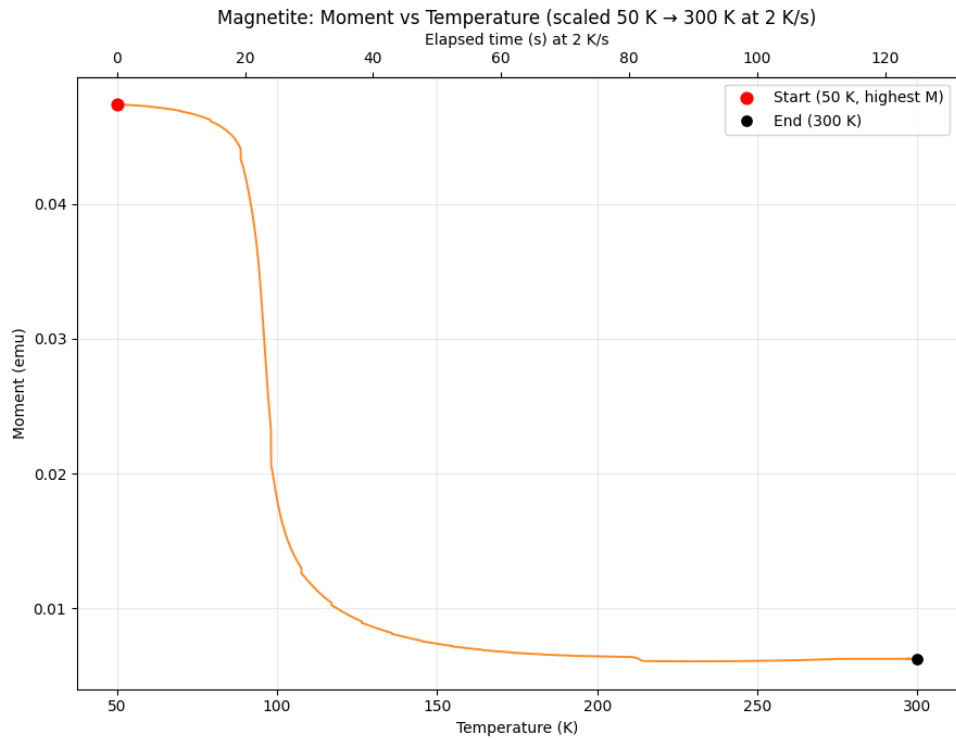


Figure 4: Magnetic moment as a function of temperature measured at an applied field of 1000 Oe. The phase transition near 119 K corresponds to the Verwey transition in magnetite.

5 Analysis

The presence of a hysteresis loop in Fig. 3 and the phase transition observed in Fig. 4 confirm that the sample exhibits the expected ferrimagnetic behavior of magnetite. However, the calculated magnetization was approximately two orders of magnitude smaller than typical literature values. The most likely causes of this discrepancy are experimental factors such as imperfect positioning of the sample within the VSM holder or insufficient centering during measurement. If the sample is not located at the optimal position relative to the pickup coils, the detected signal can be significantly reduced. Additionally, the sequence settings, particularly the use of the automatic centering routine, may have influenced the measured moment. Despite the reduced absolute scale of magnetization, the qualitative magnetic behavior of the sample was successfully measured and clearly demonstrates the characteristic properties of magnetite.

Hysteresis Loop

The magnetization as a function of applied magnetic field shows that the magnetization follows a different path during increasing and decreasing field sweeps ($0 \rightarrow 10,000 \rightarrow 10,000 \rightarrow 10,000$ Oe). This behavior forms a hysteresis loop, indicating that the magnetic state of the material depends on its magnetic history. Such hysteresis is characteristic of ferro- and ferrimagnetic materials and arises from the dynamics of magnetic domains. In particular, domain-wall motion and pinning at defects or structural inhomogeneities prevent the magnetization from immediately aligning with the applied field. As a result, the system exhibits remanent magnetization and a finite coercive field.

Verwey Transition

The magnetic moment as a function of temperature reveals a clear anomaly near approximately 120 K, which is consistent with the Verwey transition in magnetite [2]. This transition corresponds to a structural change in the crystal lattice accompanied by charge ordering of the iron ions. The change in lattice symmetry modifies the electronic and magnetic properties of the material, which can be detected in magnetic measurements. As the temperature decreases toward the transition, the magnetic moment increases due to stronger alignment of magnetic domains and reduced thermal fluctuations. The feature observed near 120 K therefore agrees with previously reported values for the Verwey transition in magnetite [1] and confirms the expected phase behavior of the sample.

6 Conclusion

In this experiment, the magnetic properties of magnetite (Fe_3O_4) were investigated using a vibrating sample magnetometer (VSM) in a Quantum Design VersaLab system. Measurements of magnetic moment as a function of applied magnetic field, $M(H)$, and temperature, $M(T)$, were used to characterize the ferrimagnetic behavior of the sample and identify the Verwey transition.

The $M(H)$ measurements produced a clear hysteresis loop, demonstrating nonzero remanent magnetization and a finite coercive field. This behavior confirms that the sample is ferrimagnetic, consistent with the known characteristic of magnetite. The shape of the hysteresis loop indicates that magnetization depends on the magnetic history of the sample, which is attributed to domain wall motion and pinning within the crystal lattice.

The $M(T)$ measurements revealed a noticeable phase transition near approximately 120 K, which is consistent with the Verwey transition in magnetite. This transition corresponds to

a structural shift of the iron sublattice that modifies the magnetic response of the material. The observation of this feature agrees with previously reported values for the Verwey transition temperature.

Although the measured magnetization was significantly smaller than previously known values, this discrepancy is most likely due to experimental factors such as imperfect centering of the sample within the pickup coils, sequencing errors, or background contributions from the sample holder. Despite these limitations, the qualitative magnetic behavior of the sample was successfully measured. The experiment therefore demonstrates the key magnetic characteristics of magnetite, including ferrimagnetic hysteresis and the presence of the Verwey phase transition.

References

- [1] E. J. W. Verwey, “Electronic conduction of magnetite (Fe_3O_4) and its transition point at low temperatures,” *Nature*, vol. 144, pp. 327–328, 1939.
- [2] F. Walz, “The Verwey transition — a topical review,” *Journal of Physics: Condensed Matter*, vol. 14, pp. R285–R340, 2002.
- [3] Quantum Design, *Vibrating Sample Magnetometer (VSM) Option User’s Manual*, Quantum Design Inc., 6th ed., 2008.
- [4] Quantum Design Educational Module, “Magnetic Characterization of Fe_3O_4 Using a VSM,” Discovery Teaching Labs.
- [5] Quantum Design, *VersaLab Physical Property Measurement System User Manual*, Quantum Design Inc.

A Python Post Processing

The Python code used to analyze the VersaLab data and generate the plots is available in the following GitHub repository:

<https://github.com/joshuaclavecillas/computational-physics/blob/74069a6bca4641059a6b46081578af501f353fc2/postprocessingversalab.ipynb>